

How to Monitor Metrics When AF_PACKET Mode is Running

Quick Answer

If you already have `run_afpacket_mode.sh` running in one terminal, simply open a second terminal and run:

```
cd /home/sujay/Programming/IDS
./monitor_metrics.sh
```

This will launch the metrics dashboard showing real-time performance statistics.

Detailed Setup

Scenario: AF_PACKET Mode Already Running

Terminal 1 - Your IDS Pipeline (already running):

```
cd /home/sujay/Programming/IDS
sudo ./run_afpacket_mode.sh
# Select option 1: Start Complete Pipeline
# This starts: Kafka → Suricata → Bridge → ML Consumer
```

Terminal 2 - Monitor Metrics:

```
cd /home/sujay/Programming/IDS
./monitor_metrics.sh
```

That's it! The dashboard will show live metrics.

Monitor Script Options

```
# Default: Launch dashboard
./monitor_metrics.sh

# Check if metrics are being generated
./monitor_metrics.sh --status

# View raw metrics in real-time (JSON format)
```

```
./monitor_metrics.sh --tail

# List all metrics files
./monitor_metrics.sh --files

# Show help
./monitor_metrics.sh --help
```

What Each Option Does

1. Dashboard (Default)

```
./monitor_metrics.sh
# or
./monitor_metrics.sh --dashboard
```

Shows:

IDS Pipeline Metrics Dashboard

LATENCY METRICS

Component: ml_consumer | Operation: ml_inference

Mean:

12.3 ms

P50:

11.2 ms

P95:

18.9 ms

P99:

25.7 ms

THROUGHPUT METRICS

Events/sec:

82.5

Total Events:

1,234

ML METRICS

Prediction Distribution:

BENIGN:

890 (89%)

DoS:

78 (8%)

PortScan:

32 (3%)

[Refreshes every 5 seconds]

2. Status Check

```
./monitor_metrics.sh --status
```

Shows:

- ✓ Which pipeline components are running
- ✓ Metrics files location and size
- ✓ Latest metrics timestamp
- ⚠ Warnings if components are missing

Example output:

```
Checking pipeline status...
✓ Kafka Bridge is running
✓ ML Consumer is running

✓ All pipeline components running!

Checking metrics files...
✓ Found 1 metrics file(s)

Metrics files:
-rw-rw-r-- 1 sujay sujay 145K Oct 26 15:30 metrics_20251026.jsonl
-rw-rw-r-- 1 sujay sujay  89K Oct 26 15:30 metrics_20251026.csv

Latest metrics file: metrics_20251026.jsonl
File size: 145K
Line count: 3456 metrics
Last modified: 2025-10-26 15:30:45
```

3. Tail Metrics

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```
./monitor_metrics.sh --tail
```

Shows: Raw JSON metrics in real-time (useful for debugging)

Example output:

```
Tailing: metrics_20251026.jsonl
(Press Ctrl+C to exit)

{
  "timestamp": 1729959045.123,
  "metric_type": "latency",
  "component": "ml_consumer",
  "operation": "ml_inference",
```

```
"latency_ms": 12.345
}
{
  "timestamp": 1729959045.456,
  "metric_type": "throughput",
  "component": "ml_consumer",
  "events_count": 100,
  "events_per_second": 82.5
}
```

4. List Files

```
./monitor_metrics.sh --files
```

Shows: All metrics files in `logs/metrics/` directory

Complete Workflow Example

Step-by-Step: Monitoring an Already Running Pipeline

Step 1: Check if pipeline is running

```
./monitor_metrics.sh --status
```

If you see:

- ✓ Components running → Good! Proceed to Step 2
- ⚠ Components missing → Start them with `run_afpacket_mode.sh`

Step 2: Launch dashboard

```
./monitor_metrics.sh
```

Step 3: Watch metrics update

- Dashboard refreshes every 5 seconds
- Shows real-time performance
- Press Ctrl+C to exit

Step 4: Analyze historical data (optional)

```
# View raw data
cat logs/metrics/metrics_$(date +%Y%m%d).jsonl | jq '.'
```

```
# Open in spreadsheet
libreoffice logs/metrics/metrics_$(date +%Y%m%d).csv
```

Terminal Layout Recommendation

2-Terminal Setup (Simple)

Terminal 1: Pipeline	Terminal 2: Dashboard
\$ sudo ./run_afpacket_mode.sh	\$./monitor_metrics.sh
[INFO] Processing...	▄▄ Metrics Dashboard
[INFO] Prediction: BENIGN	Latency: 12ms (P95)
	Throughput: 82 evt/s

3-Terminal Setup (Advanced)

Terminal 1: Pipeline	Terminal 2: Dashboard	Terminal 3: Logs
\$ sudo ./run_afpacket_mode	\$./monitor_metrics.sh	\$ tail -f logs/ml/*.log
[Running...]	▄▄ Dashboard	[INFO] Events...

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Troubleshooting

Problem: Dashboard shows "No metrics files found"

Cause: Pipeline hasn't generated metrics yet

Solution:

```
# 1. Check if pipeline is running
./monitor_metrics.sh --status

# 2. If components are running, wait 10-30 seconds
```

```
# Metrics are flushed every 30 seconds

# 3. Check again
./monitor_metrics.sh --status

# 4. If still no metrics, check if MetricsLogger is enabled in code
grep -r "MetricsLogger" dpdk_suricata_ml_pipeline/src/
```

Problem: "Pipeline not running" warning

Cause: `run_afpacket_mode.sh` not started yet

Solution:

```
# Terminal 1: Start pipeline first
sudo ./run_afpacket_mode.sh
# Select option 1: Start Complete Pipeline

# Terminal 2: Then monitor
./monitor_metrics.sh
```

Problem: Dashboard shows stale data

Cause: Pipeline stopped, dashboard reading old files

Solution:

```
# Check if pipeline is still running
ps aux | grep -E 'suricata|bridge|consumer'

# If not running, restart:
sudo ./run_afpacket_mode.sh
```

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Integration with DPDK Mode

The monitoring script works **identically** with DPDK mode:

Terminal 1 - DPDK Pipeline:

```
sudo ./run_dpdk_mode.sh
```

Terminal 2 - Monitor (same command):

```
./monitor_metrics.sh
```

The script auto-detects which mode is running and displays appropriate metrics.

Key Metrics to Watch

For AF_PACKET Mode

Good Performance:

- **P95 Latency:** < 100ms end-to-end
- **Throughput:** 50-200 events/sec
- **CPU:** 50-80%
- **Confidence:** > 0.8

Warning Signs:

- P99 > 500ms → Investigate bottlenecks
 - Throughput < 10/s → Check for errors
 - Many errors → Check logs
 - Low confidence < 0.5 → Model quality issue
-

Quick Reference

```
# Most common usage
./monitor_metrics.sh                # Launch dashboard

# Check if working
./monitor_metrics.sh --status       # Status check

# Debug
./monitor_metrics.sh --tail         # Raw metrics
tail -f logs/ml/ml_consumer.log     # ML consumer logs

# Analysis
cat logs/metrics/metrics_*.jsonl | jq '.'          # Pretty JSON
libreoffice logs/metrics/metrics_$(date +%Y%m%d).csv # Spreadsheet
```

Summary

Simple answer: Open a second terminal and run `./monitor_metrics.sh`

What it does:

- ✓ Checks if pipeline is running

- ✓ Verifies metrics are being generated
- ✓ Launches real-time dashboard
- ✓ Updates every 5 seconds
- ✓ Works with both AF_PACKET and DPDK modes

Files created:

- `monitor_metrics.sh` - Monitoring script (executable)
- `logs/metrics/metrics_YYYYMMDD.jsonl` - Metrics data
- `logs/metrics/metrics_YYYYMMDD.csv` - Spreadsheet format